U.S. Digital Health Unchained: A Manifesto to Liberate Patients and Providers in the Quantum Era

Response to CMS-0042-NC: "Health Technology Ecosystem"

Submitted to:
U.S. Department of Health and Human Services (HHS)
Centers for Medicare & Medicaid Services (CMS)
Assistant Secretary for Technology Policy (ASTP), Office of the National Coordinator for Health IT (ONC)

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Submitted via: www.regulations.gov
Docket ID: CMS-2025-0050

Date of Submission: June 11, 2025

Respectfully submitted,

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This response includes structured recommendations aligned to all stakeholder categories outlined in the RFI (Patients, Providers, Technology Developers, and Value-Based Organizations), with section-level mapping and RFI code tagging throughout.

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I. Executive Summary

The U.S. healthcare system stands at a once-in-a-century inflection point. What lies ahead is not another framework or regulation, but a chance to rewrite the operating code of American medicine. This response offers that code.

Salphaware, LLC is honored to respond to this Request for Information (RFI) issued by the Centers for Medicare & Medicaid Services (CMS) and the Office of the National Coordinator for Health Information Technology (ASTP/ONC). As a veteran-owned health technology company founded by professionals with deep technical, clinical, and regulatory expertise, we bring a vision rooted in service, shaped by frontline insight, and governed by the conviction that digital health must now serve the nation rather than the market alone.

Patient health data has not simply been siloed. It has been rounded up, fenced off, and made subject to proprietary captivity—no longer a public good, but a controlled commodity. Billions of dollars and countless workarounds have failed to produce the interoperability Americans were promised. What is needed now is neither another fix nor interface, but a unified digital fortress, engineered for national coordination and local execution.

This document architects a bold yet grounded redesign of the U.S. health information system. At its center is a single, urgent principle: health data is a national asset—and must be protected, structured, and activated as such. Our proposal untethers legacy concepts through a National Health Data Repository, atomic-level data architecture, universal identity layers for patients and entities, cybersecurity to rival the intelligence sector, a harmonized public health empire, and a single federated patient portal.

Together, these elements herald the quantum era of U.S. healthcare—where emergency response is real-time, fraud is intercepted before it spreads, AI-enabled care is the norm, and the ecosystem's reach—urban, rural, tribal, territory—manifests as equity.

Salphaware's flagship offering, *Sepsis Studio*, automates CMS SEP-1 abstraction, retroactive DRG appeals, and real-time compliance nudging—fully aligned to CMS' burden-reduction priorities under Section 4001 of the 21st Century Cures Act. The proof-of-concept demonstrates that what has long been deemed complex is, in fact, achievable. But sepsis is only the start. Our broader roadmap is guided by the Trilateral Value Proposition—a framework developed by Salphaware to center U.S. health system reform around value for three core stakeholders: the patient, the provider, and the federal government. Every technical proposal within this document answers to all three.

What follows is defensible in design, transformative in consequence, and ethically courageous. We submit this not merely as a response—but as a call to collaborate. The opportunity before us is nothing short of historic: to unchain American healthcare from its legacy constraints, and to build a system worthy of its people—secure by design, self-aware in operation, and unrelenting in its service to life.

II. Contract Innovation Through the Trilateral Value Proposition Framework

This section addresses the following CMS RFI questions:
PR-2, PR-8, TD-1, VB-7

A. The Trilateral Value Proposition: A Contractual Framework for Serving Patients, Providers, and Government

The U.S. health system is not failing from lack of effort—it is failing from misalignment. Policies serve reporting requirements instead of people. Contracts reward efficiency without demanding impact. Providers are burdened. Patients are fragmented. And the very tools built to transform healthcare have too often replicated the paperwork logic they were meant to eliminate.

But this is our inflection point. We can all be titans of long-overdue transformation if we align on philosophy first and negotiate later.

At the heart of this response is a single, unapologetic principle: value must flow in at least three directions—to the patient, to the provider, and to the federal system. This is the Trilateral Value Proposition (TVP): a practical, measurable, and morally grounded framework pioneered by Salphaware to re-anchor federal healthcare innovation to what matters more than any regulation or budget line.

The TVP is not just a philosophy; it should be hard-coded into procurement mechanics. It reframes contracts not as checklists, but as instruments of design and repair. And it insists on a symmetrical definition of success:

- If a tool improves government efficiency but increases provider burnout, it fails.
- If it supports providers but renders patients invisible, it fails.
- If it serves patients but is unscalable or inequitable, it fails *[PR-2]*.

Adopting the TVP as a contracting standard multiplies the genomic depth of CMMI into every transaction, converting human-scale innovation into minimum viable product and with maximum systemic return.

The TVP asks a single litmus-test question: Does this solution deliver mission-aligned value to all three?

- To patients: It means safer, faster, more coherent care—where continuity follows the person, not the institution. It means deterioration caught earlier, referrals executed smoother, and in critical moments like sepsis, outcomes determined—not just documented.

- To providers: It means less documentation, less duplication, less drag. More clarity. More time. More trust. TVP-aligned tools reinforce clinical reasoning rather than replace it—because digital health should elevate clinicians, not override them *[PR-8].*
- To CMS: It means structured, clinically grounded data—ready to support accurate payment, real-time regulatory oversight, public health surveillance, and the evolution to scalable, self-improving systems.

We urge CMS to elevate the TVP as a baseline for all future contracts, grants, and initiatives. This shift would not only modernize procurement; it would reorient the federal innovation ecosystem itself. Imagine if every submission, every solution, every vendor had to prove not just utility but mutuality. Imagine a future where value to the patient and provider is no longer a bonus; it is the requirement.

To accelerate this shift, CMS could:

- Integrate TVP scoring into Medicare Shared Savings Program (MSSP) quality metrics
- Prioritize vendors and proposals that deliver demonstrable tri-directional value
- Offer innovation grants or payment multipliers for tools that fulfill TVP criteria—especially in high-burden conditions or across shared-risk models like Accountable Care Organizations (ACOs) and Alternative Payment Models (APMs) *[VB-7]*

The message is simple but profound: innovation is not decoration; it is obligation. And contracts are no longer forms; they are transformative blueprints.

Somewhere in the shared complexity of our system, the patient has been lost. Patients do not experience healthcare as line items, metrics, or dashboards. They experience it as deeply personal and sometimes terrifying moments that determine whether they live, suffer, or heal.

The Trilateral Value Proposition is how we honor those moments, and how we finally build a system that sees them, hears them, and answers.

B. De-Risking Hospital Participation in Innovation Pilots

Innovation cannot transform the system if it is only tested in its wealthiest corners.

Hospitals are the proving grounds of national health reform, but without structured support, too many are left on the sidelines. Safety-net, rural, and lower-resourced institutions often face overwhelming legal, technical, and operational hurdles when invited to test early-stage solutions. The result is a system where innovation scales first to the advantaged, and slowly—if ever—to those who need it most.

CMS can and should reverse this trend.

We recommend the creation of structured, de-risking incentives for hospital participation in CMS-aligned innovation pilots *[TD-1]*. These mechanisms would recognize hospitals not as passive endpoints, but as active collaborators in modernizing the national health infrastructure.

Policy options may include:

- Pilot-site stipends or technical assistance grants to offset implementation and staffing costs;
- Regulatory flexibilities under the Quality Payment Program (QPP) to reduce administrative load;
- Scoring bonuses within MSSP or future value-based arrangements for approved pilot participation;
- Shared-risk or liability safe harbors for institutions testing certified tools in CMS-authorized pilots;
- National early-adopter recognition for hospitals contributing real-world validation across diverse care environments.

These supports could be coupled with CEHRT flexibility provisions under *VB-10* to ensure that participation in digital innovation pilots does not penalize or exclude hospitals based on certification burdens they are structurally less able to meet.

Hospitals should not have to bear the cost of transformation alone. By lowering the risk and raising the reward, CMS can catalyze a broader, faster, and more equitable deployment of tools like *Sepsis Studio*—tools designed not for the margins, but for the mission. These incentives would also operationalize the TVP: delivering earlier access to innovation for patients, reducing strain on providers, and improving implementation fidelity for policymakers.

De-risking is not a subsidy. It is a strategic accelerant. It signals that innovation is no longer something hospitals are asked to try, but rather something they are equipped to lead.

C. Prioritizing Veteran-Owned Small Businesses in Innovation Contracts

Veteran-Owned Small Businesses (VOSBs) bring a rare and underutilized asset to federal healthcare transformation: mission-driven expertise grounded in lived service. Many VOSBs are led by professionals who have navigated the very systems now targeted for modernization—from CMS quality programs and public health infrastructure to the military's own complex care delivery networks.

This is not about inclusion—it is about performance.

Even though CMS and ASTP/ONC did not specifically solicit feedback on this point in the RFI, we believe the role of VOSBs in advancing America's health technology

infrastructure should not remain unspoken. Veteran leadership is a public asset, not a footnote.

We recommend that CMS adopt explicit contract scoring preferences for qualified VOSBs engaged in health IT, burden reduction, and national modernization efforts. These preferences could mirror those outlined in FAR Subpart 19, with tailored emphasis for VOSBs aligned to CMS strategic priorities.

Doing so would:

- Expand the talent pool of federal innovation partners to include those with both technical skill and civic grounding;
- Advance Executive Order 13985 through measurable, mission-aligned participation;
- Signal to the market that veteran-owned does not mean auxiliary—it means embedded in mission-critical outcomes.

Salphaware is a majority veteran-owned small business advancing through SBA's VetCert process. Its leadership reflects the core VOSB advantage: service-oriented innovation, systems-level understanding, and a commitment to public good. Like many others, it is not asking for special treatment—it is asking for a seat at the table it helped set.

III. National Health Technology Infrastructure and Cybersecurity Strategy

This section addresses the following CMS RFI questions:
PC-1, PC-2, PC-3, PC-5, PC-6, PC-8, PC-11, PC-12, PC-13, PC-14, PR-1, PR-3, PR-4, PR-7, PR-9, PR-10, PR-12, PA-1, PA-2, PA-3, TD-1, TD-2, TD-3, TD-12, TD-13, TD-15, TD-16, TD-17, TD-18, TD-19, VB-2, VB-4, VB-12, VB-13, VB-14

The barrier to an intelligent, resilient health system is no longer technological—it is structural. Artificial intelligence, machine learning, and advanced analytics are already capable of supporting predictive care, real-time surveillance, and automation of clinical and regulatory tasks. What stands in the way is not model capacity. Instead, the barrier is the absence of clean, accessible, secure, and interoperable data *[PC-8]*.

Today, U.S. health data systems remain shaped by outdated billing structures, proprietary vendor silos, and encounter-based logic. These architectures serve internal accounting—not longitudinal care, not public health coordination, and not national resilience. The result is fragmentation, brittleness, and an inability to respond coherently to threats ranging from sepsis to cyberattacks to bioterrorism *[PC-11]*.

A durable future requires more than policy tweaks. It requires the re-architecture of U.S. health data as critical infrastructure. This transformation must be led by federal agencies—CMS, and ONC among them—with the same foresight and rigor applied to energy, transportation, and national defense.

To this end, Salphaware proposes and supports a triad of interlocking federal constructs:

1. The National Health Data Repository (NHDR): A federally governed system of record that supports real-time analytics, care continuity, and public health mobilization
2. The Atomic Health Data Architecture (AHDA): A granular, extensible, time-stamped data structure designed for AI, transparency, and reuse across the continuum of care
3. The National Health Citizen and Entity Identifiers (NHCI/NHEI): A persistent digital identity framework that links patients and provider organizations across space, time, and systems

Each component reflects a simple truth: data infrastructure determines system capability. Without transformation at this foundational level, automation remains constrained, AI remains limited, and public trust remains undermined.

This is not a speculative vision. It is an executable blueprint. The most urgent innovation is not artificial intelligence. It is architectural intelligence.

A secure, longitudinal, interoperable data infrastructure is the necessary substrate for every other reform goal: quality improvement, public health surveillance, health equity, cost transparency, emergency response, and patient empowerment. Without it, the promise of intelligent health systems remains unrealized.

Salphaware affirms this vision as a veteran-led organization grounded in service, clinical expertise, and federal contracting experience. Our position is clear: the future of healthcare depends not on what machines can learn, but on what systems let them see.

This section presents the case—and the path—for building the infrastructure the nation needs now.

A. National Health Identity Standards: NHCI and NHEI for Trust, Continuity, and Coordination

Every system of national significance—whether Social Security, passport control, or tax administration—relies on reliable, persistent identifiers. Healthcare remains a glaring exception. The absence of standardized, interoperable identity structures undermines continuity, coordination, and accountability across every node of the delivery system *[PC-8]*.

We propose two foundational identifiers for U.S. healthcare data infrastructure:

- The National Health Citizen Identifier (NHCI): a persistent, context-agnostic patient identifier that enables longitudinal continuity across time, geography, and settings of care
- The National Health Entity Identifier (NHEI): a unique identifier for each healthcare-delivering organization—EMS agencies, primary care clinics, dialysis centers, hospitals, nursing homes, community health centers, and others—regardless of size or affiliation

Together, these identifiers provide a clean, scalable scaffolding for trustworthy, cross-continuum data exchange *[PR-3]*. They enable traceability without reliance on local medical record numbers, payer-issued IDs, or institution-specific constructs. They support dynamic linkage across siloed EHRs, enabling auditability, privacy segmentation, and equitable participation in care coordination and public health response *[PR-7]*.

The precedent is clear. The U.S. Social Security Number (SSN), despite its limitations, fundamentally changed how federal systems track citizen activity, benefits, and entitlements. The National Provider Identifier (NPI), introduced in the mid-2000s, created a universal reference model across public and private healthcare. A hyphenated alphanumeric NHCI—human-readable, machine-parseable, and format-stable—would mirror these proven models. The Veterans Affairs Integration Control Number (VA ICN) and the Department of Defense Electronic Data Interchange Personal Identifier (DoD EDIPI) have demonstrated scalable, durable use across large federated systems without meaningful architectural criticism.

NHCI could also be leveraged as the foundational anchor for a national patient-facing digital credential—enabling alignment with platforms like Login.gov or ID.me to simplify access for patients while reducing administrative burden for providers *[PR-9]*.

This approach also aligns with ONC's encouragement of persistent digital identity frameworks as foundational to national interoperability *[PC-14]*. Digital identity constructs like NHCI and NHEI directly support NIST 800-63-3-aligned frameworks, enabling implementation of standards such as IAL2/AAL2 credentialing and OpenID Connect protocols—fortifying both cybersecurity posture and interoperability readiness *[TD-3]*. That same logic—clarity, trust, interoperability—now compels a national identity model for both patients and provider entities.

Crucially, the NHCI and NHEI are not surveillance tools—they are structural tools. Their purpose is not to centralize data, but to normalize, align, and secure it across a federated health network. Their use can be governed by purpose-based access rules, audit trails, and data minimization safeguards—fully aligned with the HIPAA Privacy Rule, ONC information blocking exceptions, and federal zero trust architecture guidance.

By pairing NHCI and NHEI standards with an atomic data model, CMS can radically strengthen the fidelity, trustworthiness, and usability of health data for care delivery, digital quality measurement, health equity tracking, and AI-powered analytics. This foundational standardization would also reduce the complexity and friction that often undermine provider participation in value-based arrangements, especially those requiring multi-source data aggregation and attribution fidelity *[VB-13]*. These identifiers provide the anchoring logic required to implement atomic data structures at national scale—as demonstrated in the tables and architecture introduced in the following subsection.

Importantly, NHCI could also streamline patient authentication across payer systems, encouraging broader adoption of shared credentials like Login.gov or ID.me to reduce access friction and member confusion *[PA-3]*.

A modern health system cannot function without persistent identity logic. The time has come to define and deploy it.

B. Atomic Health Data Architecture: Enabling Structured Intelligence

Every high-functioning national system—whether financial, transportation, or defense—relies on data structures engineered for scale, precision, and resilience. Healthcare should be no different. Yet today's health data models remain encounter-bound, non-normalized, and vendor-defined—optimized for billing, not insight. Salphaware rejects this architectural confinement. We propose an Atomic Health Data Architecture (AHDA) to correct this misalignment with a patient-centric, longitudinal foundation built for intelligence, precision, and durability.

It is a fact that patient access barriers are engineered, often unintentionally, by the structure of health data itself. The system's architectural rigidity and self-imposed

constraints on data evolution entrench fragmentation. AHDA reverses this logic. It allows records to follow the patient, not the silo. It redefines data as a longitudinal asset—patient-loyal across time, geography, and platform.

While Salphaware's current implementation leverages SQL for backward-compatible abstraction, our roadmap includes full compliance with CEHRT requirements and certified FHIR-based APIs per the ONC Health IT Certification Program. These certified pathways will be prioritized during deployment to ensure full interoperability with CMS expectations.

The cornerstone of resilient health infrastructure is a normalized, extensible data model that enables atomic precision and semantic integrity. AHDA implements variable-specific longitudinal tables—each designed to accommodate temporal granularity, source provenance, and cross-entity normalization.

Unlike aggregation-based models such as FHIR—which consolidate data post hoc and suffer from semantic drift, resource bloat, and inconsistent implementation—AHDA is architected for reuse, automation, and structural clarity. This allows integration of AI, risk stratification, real-time quality measurement, and digital nudging into APM workflows with minimal reengineering *[VB-2]*. By decoupling the data model from vendor constraints and encounter logic, AHDA enables scalable digital quality measurement, real-time compliance nudging, and AI-powered abstraction.

The following illustrative tables present a simplified model for a single clinical variable—heart rate—demonstrating how each observation is independently recorded across time, systems, and care settings. Note the presence of NHCI and NHEI fields as core components of extensibility and traceability.

Figure 1 – Example atomic variable table: single patient, single hospital length of stay

NHCI	Timestamp	Variable	Value	Unit	Source	Entity	NHEI
A17X-4406-US11	2026-01-06 13:13	Heart Rate	98	BPM	Monitor	Hospital	H330201
A17X-4406-US11	2026-01-06 13:28	Heart Rate	102	BPM	RN	Hospital	H330201
A17X-4406-US11	2026-01-06 13:43	Heart Rate	96	BPM	Monitor	Hospital	H330201
A17X-4406-US11	2026-01-06 13:58	Heart Rate	100	BPM	RRT	Hospital	H330201
A17X-4406-US11	2026-01-06 14:13	Heart Rate	99	BPM	MD	Hospital	H330201

Figure 2 – Example atomic variable table showing extensible architecture

NHCI	Timestamp	Variable	Value	Unit	Source	Entity	NHEI
A17X-4406-US11	2026-01-03 09:30	Heart Rate	88	BPM	RN	Primary	PC1234
A17X-4406-US11	2026-01-04 11:00	Heart Rate	90	BPM	MD	Specialty	SC2222
A17X-4406-US11	2026-01-06 12:45	Heart Rate	95	BPM	EMT	EMS	EMS987
A17X-4406-US11	2026-01-06 13:13	Heart Rate	98	BPM	Monitor	Hospital	H330201
A17X-4406-US11	2026-01-06 13:28	Heart Rate	102	BPM	RN	Hospital	H330201
A17X-4406-US11	2026-01-06 13:43	Heart Rate	96	BPM	Monitor	Hospital	H330201
A17X-4406-US11	2026-01-06 13:50	Heart Rate	92	BPM	RN	Dialysis	DIAL110
A17X-4406-US11	2026-01-06 13:58	Heart Rate	100	BPM	RRT	Hospital	H330201
A17X-4406-US11	2026-01-06 14:13	Heart Rate	99	BPM	MD	Hospital	H330201
A17X-4406-US11	2026-01-07 08:45	Heart Rate	87	BPM	RN	CHC	CHC5566
A17X-4406-US11	2026-01-10 10:15	Heart Rate	85	BPM	RN	SNF	NH3344

Each table is variable-specific and structured around a primary key that includes patient identity (NHCI), temporal fidelity (Timestamp), and contextual metadata (Source, Entity, NHEI). This identity-driven schema supports credential-based interoperability frameworks, including those aligned with NIST 800-63-3 and OpenID Connect protocols for secure, federated transmission [*TD-3*]. These structured data types—time-resolved vitals, identity anchors, and care context—are foundational for longitudinal attribution, risk modeling, and quality analytics in value-based care [*VB-4*]. Together, they enable traceability across EMS, primary care, specialty care, hospitals, dialysis centers, community health centers, and nursing homes—supporting normalization across the entire continuum of care.

AHDA enables longitudinal normalization, precision analytics, and secure, federated transmission. Its extensibility supports integration with non-CMS data networks where fidelity, traceability, and normalized identity models meet federal criteria for endorsement *[TD-12]*. It directly supports CMS initiatives such as digital quality measurement (dQM), data segmentation for privacy (DS4P), and the vision for clean, AI-ready datasets aligned with the USCDI v3 framework.

Bleeding-edge data science principles cannot remain an afterthought in CMS' immediate-term vision—especially given what is already possible, with or without AI. AHDA provides an operational pathway to meet these possibilities with minimal provider-side lift, lowering implementation burden while increasing analytical return *[VB-13]*. At the 2025 State HIT Connect Summit, one theme was clear: *the industry remains far from achieving clean, computable data*. Salphaware addresses this gap with AHDA—a future-ready model built for automation, extensibility, and certified interoperability.

This work aligns with ONC's regulatory framework under 45 CFR § 170 Subpart C, which defines certification criteria for interoperability, clinical data exchange, and support for patient access. It anticipates and supports USCDI v3 adoption, standardizing core data elements for nationwide interoperability and public health exchange.

Whereas FHIR aggregates inconsistent records in complex formats, AHDA restructures data at its core. It achieves maximum granularity in the cleanest, most portable structure possible. AHDA does not merely comply with interoperability standards—it redefines them.

And in doing so, it delivers the foundational logic required for reusable, model-ready, cross-domain health intelligence at national scale.

C. National Health Data Repository: Infrastructure-Grade Intelligence for the Nation

Most Americans have already received the letter.

A breach. A compromise. A warning that their medical information, Social Security number, or insurance data has been exposed—through no fault of their own. From ransomware attacks on major hospital systems to third-party vendor hacks affecting millions, the message is always the same: your most private information is now in someone else's hands.

This is not just an IT problem. It is a national vulnerability—one that affects citizens directly and viscerally. In the modern age, personal health data is personal safety. And the United States has no meaningful system to protect it.

That must change.

The National Health Data Repository (NHDR) is the infrastructure-grade solution to this national failing. It is a secure, federally governed system of record that decouples core health data from the fragmented, vendor-defined platforms that currently dominate U.S. healthcare. It transforms health data into a national asset—secured with quantum-resilient safeguards, protected by clear access policies, and made available to patients, providers, and public health officials through trusted, certified interfaces that expose full EHI access via APIs for patients, providers, and payers—accelerating downstream innovation in the spirit of CMS's Blue Button 2.0 and Data at the Point of Care models *[PA-2]*. These interfaces would expose complete EHI via secure, standardized APIs—unleashing innovation while resolving the limitations of partial access frameworks such as USCDI alone *[TD-13]*. Also, this includes identity-verified access aligned with NIST 800-63-3 digital credentialing frameworks to ensure secure interoperability across provider systems *[PR-10]*.

This is not about centralizing all health data in a monolith. It is about national resilience. The NHDR replaces point-to-point connectivity with a scalable, shared infrastructure—reducing redundant integrations, enabling reciprocal access, and overcoming the fragmentation seen in current federated architectures, including TEFCA *[TD-16, PA-1]*. Rather than replacing every EHR, it establishes a persistent backbone across life stages, provider networks, and disaster scenarios. It is the modern equivalent of the interstate system or the electrical grid: not a luxury, but a public good foundational to national function.

For patients, it means never starting over. No more recounting medication lists to every new specialist. No more missing allergy documentation. No more fractured records scattered across five logins, three hospitals, and two state lines *[PC-8]*. This model presumes identity continuity through NIST-aligned credentials and OpenID Connect integration—ensuring verifiable access across disparate systems while reducing login fatigue and information silos *[TD-3]*. A secure, persistent record—tied to a verified health identity—follows them wherever care is needed, and supports payer-facing access via credentialing platforms like Login.gov or ID.me *[PA-3]*.

For providers, it means access to clean, structured, longitudinal data—without hours spent hunting through scanned PDFs or manually reconciling outside records. For nurses,

physicians, EMS professionals, and pharmacists alike, the NHDR becomes a trusted baseline for informed decisions, enabled through verified identity protocols that streamline data access while mitigating credentialing and reconciliation burdens *[PR-1, PR-10, VB-14]*. This level of data quality and consistency enables more reliable performance measurement, streamlines compliance tasks, and supports continuous improvement in value-based care delivery *[VB-13]*.

For public health agencies and CMS, it means real-time situational awareness. It ends the era of retrospective extracts, delayed case reporting, and black-box datasets. With the NHDR in place, fraud detection, outbreak monitoring, value-based payment, and emergency coordination are no longer siloed—they are synchronized.

And for the nation, it means security. Today, health data is protected—or not—by a patchwork of private vendors with competing interests, uneven protocols, and no unified defense. The NHDR replaces that friability with standards. With auditability. With a data architecture built to withstand cyber threats, foreign interference, and cascading failures. It also relieves hospitals and health systems of the impossible burden—and legal liability—of securing digital infrastructure they neither built nor control. These persistent gaps in readiness and response capability are exacerbated by fragmented access rules, unpredictable interfaces, and information blocking behaviors that disincentivize timely sharing *[TD-18]*. In doing so, it addresses structural gaps that current information blocking exceptions have failed to remedy—enabling true access, use, and innovation *[PR-12]*. No other industry demands that frontline service providers double as national cybersecurity guardians. The NHDR ends this untenable expectation and replaces it with a rational, federally secured backbone *[PR-7]*.

Importantly, this model does not stifle innovation. It unleashes it. By decoupling data access from proprietary systems, the NHDR allows any certified application—AI tools, clinical dashboards, decision support systems, public health portals—to operate on a shared substrate *[VB-12]*. API-enabled access to complete longitudinal EHI supports more powerful and personalized applications, although doing so requires careful balancing of data utility, privacy, and governance *[TD-13]*. Developers no longer need special integrations. Hospitals no longer need workarounds. The floor is raised for everyone.

The NHDR follows the pattern of other national systems that Americans already rely on every day:

- The Social Security Administration enables continuity of benefits across agencies and decades.
- The National Weather Service powers commercial apps and emergency alerts alike from a shared data layer.
- The Federal Reserve supports commercial banking from a common, secure transmission infrastructure.

Healthcare is the only major domain without such a backbone. And it shows.

By establishing the NHDR, CMS and its federal partners would do more than modernize interoperability. They would restore trust. They would also ensure the viability of national data sharing by anchoring both supply (infrastructure, access, APIs) and demand (certified applications, public health use, patient engagement) in a coherent backbone designed for measurable usage and impact *[TD-17]*. It would also clarify accountability: by decoupling access from vendor discretion and enabling patients to trace where and how their data flows, the NHDR could make information blocking easier to detect and report—amplifying the purpose of the ONC's Information Blocking Portal and similar transparency mechanisms *[PC-13]*. They would give every American a rational answer to the question: "Who protects my health data?" And that answer, for the first time, would be their government—not a vendor.

This is the only viable path forward for a coherent, intelligent, and secure national health system. EHRs will remain—but as interfaces, their value defined by provider usability and system intelligence. AI will proliferate—but grounded in reliable, cross-service, cross-continuum national data. Patients will heal—not repeat their stories, not redo their labs, and not serve as secretaries of their own survival through password resets and portal fatigue. And the United States will no longer treat its citizens' most sensitive data as a commercial afterthought, but as the infrastructure-grade public trust it has always deserved to be.

D. National Security-Grade Technology Infrastructure: Moving Beyond FHIR and Toward Post-Quantum Cybersecurity

The health data of 330 million Americans is now a national security asset—and a target. Breach notifications are no longer rare or shocking; they are routine. Health systems have been paralyzed by ransomware. Patient records have been sold on the dark web. Insurers, labs, state agencies, and technology vendors alike have all issued the same message to the public: "We regret to inform you…"

This ends now.

The current U.S. health IT architecture was not built to defend a national asset. It was built to move billing codes and comply with Meaningful Use. Its foundational protocols—especially FHIR—were designed to facilitate inter-vendor communication, not to secure longitudinal identity or ensure cyber-resilience. While FHIR has advanced interoperability in limited contexts, its structural limitations now pose systemic risk when applied to infrastructure-scale demands.

FHIR's implementation model violates several foundational principles of durable system design. It duplicates metadata and structure across resources, violating the DRY (Don't Repeat Yourself) principle. It encodes context into transport formats instead of grounding it in normalized data layers, leading to semantic drift—where terms and meanings quietly shift over time, eroding interoperability and data fidelity. It wraps core data in vendor-variable resources that break under versioning. And it relies on ever-changing implementation guides that differ by use case, geography, and vendor. Most critically, it

presumes that EHR vendors—rather than a federal backbone—should remain the de facto custodians of the nation's health data. At Salphaware, we imagine a future in which the EHR oligopoly no longer controls health data transferability through self-serving obstinance.

FHIR is not a malicious standard, but it is the wrong substrate for national resilience. Its use case should remain narrow: interfacing legacy vendors and transitional APIs. While FHIR remains valuable for legacy interoperability, its architectural underpinnings limit its viability as a national-scale substrate. Even within its intended role, FHIR is compute-heavy, operationally complex, and costly to scale. It forces hospitals and developers to interpret volatile implementation guides, manage constant version churn, and build or license bespoke adapters for data normalization—each adding friction, technical debt, and downstream risk. These limitations are well known, even within ONC FAST and Argonaut forums, and have been acknowledged by FHIR developers themselves. Even where bulk FHIR APIs aim to improve data flow, their operational cost, infrastructure burden, and security surface area expand significantly—posing financial and technical viability concerns for both providers and vendors [TD-15]. Real-time use cases collapse under the weight. Edge devices cannot process FHIR payloads without substantial middleware. These are not inconveniences. They are architectural disqualifications.

Although regrettably technical, several proven or emerging architectures offer transmission models that are materially superior to FHIR, with real-world applicability. They warrant inclusion here to ensure technical audiences are not preserving FHIR out of familiarity at the expense of national cybersecurity. For example, DDS (Data Distribution Service), deployed in aerospace and defense, supports real-time, QoS-governed, publish-subscribe messaging ideal for latency-sensitive clinical scenarios. gRPC with Protocol Buffers, widely used at Google and in Kubernetes, delivers compact, strongly typed, high-throughput RPCs well-suited for backend health data exchange. ZeroMQ and NATS, foundational in high-frequency trading and IoT, enable lightweight, decentralized messaging across federated networks—with ZeroMQ operating brokerlessly and NATS leveraging a minimal server-based architecture. IPFS (InterPlanetary File System), used in decentralized research and provenance-focused archives, enables tamper-evident, content-addressed storage for immutable health records. Matrix Protocol, adopted by the French government, supports federated, end-to-end encrypted messaging applicable to secure inter-organizational and public health communications. DAG-based ledgers, including IOTA and Holochain, support asynchronous, provenance-preserving record flows and are actively explored for identity and audit chains. Post-quantum cryptographic protocols, including NIST-selected finalists such as Kyber and Dilithium, are already being piloted in international health IT systems to harden transmissions against quantum-era threats. These architectures—individually or in hybrid—represent viable successors to REST-based paradigms and offer a forward-compatible foundation for a resilient, scalable, and secure national health data infrastructure.

These transmission architectures complement or surpass FHIR's REST-based exchange by addressing what FHIR cannot: real-time delivery, strong typing, verifiable integrity,

and defense-grade resilience—without requiring abandonment of FHIR's data schemas where appropriate.

In short, FHIR's use case should remain transitional and narrow. Its architecture was designed for cooperation, not defense. It cannot meet the performance, security, or adaptability requirements of a zero-trust, post-quantum infrastructure. Continuing to scale it beyond its fit-for-purpose boundaries risks embedding technical debt into the very foundation of national health security.

Atomic health data shipping—potentially under quantum-secure, entangled-node architectures—is not speculative fantasy. It is a rational extension of current national security needs and edge-case hospital realities where interoperability must be instant, auditable, and functionally fail-proof. FHIR was not designed for—and cannot easily adapt to—the demands of a national security-grade architecture. AHDA is.

The path forward requires more than patches. It requires realignment with foundational truths:

1. Data is now infrastructure. Like energy or finance, it must be governed, standardized, and secured as a national asset—not a fragmented commercial product.
2. Security is not optional. Post-quantum cryptography and zero-trust enforcement are no longer theoretical. Quantum computing will not hack systems one by one—it will brute-force decrypt at scale, collapsing decades of cryptographic trust in a single generation.
3. Trust is structural. Secure systems are not built with firewalls and policy statements. They are built with architectures that eliminate compromise, enforce transparent auditability, and align technical design with democratic values.

Salphaware supports FHIR for near-term compliance, but we urge CMS and ONC to begin the transition toward next-generation interoperability protocols. Rather than proposing a false equivalence between interoperability protocols like FHIR and system-level programming languages, we draw attention to the fact that defense and intelligence communities have begun reevaluating foundational technology stacks—including language-level security models—in pursuit of hardened architectures. Rust, for example, is gaining favor due to its compile-time memory safety, low-level control, and resistance to class-based vulnerabilities. These features enable secure, performant systems even in adversarial or resource-constrained environments—essential for healthcare infrastructure operating at national scale.

CMS and ASTP/ONC are humbly advised to:

- Reference the initiation of the Translating All C to Rust (TRACTOR) program by the Defense Advanced Research Projects Agency (DARPA), which refactors legacy codebases into memory-safe implementations using Rust for security and efficiency.

- Heed the guidance of the National Security Agency (NSA) in recommending Rust for secure, memory-safe system architectures.
- Explore architectures inspired by intelligence-grade, mission-critical transmission models that optimize for speed, integrity, and computational minimalism.

Simple operational economy is factor here, too. As a matter of business cost mitigation, Salphaware is considering cutting-edge methods for data storage and transmission—not only programming language alternatives, but also mathematical transformations, and emerging paradigms such as quantum entanglement. While it may seem futuristic and certainly not yet production-ready for healthcare, entanglement-based communication models are being prototyped by DARPA, NASA, IBM's Quantum Network, AWS Braket, and formalized in the U.S. Department of Energy's blueprint for a national quantum internet—reinforcing the need for CMS to track these trajectories.

CMS needs bold, tech-fearless vendors asking these questions now—not later—to anticipate what comes next rather than extend what legacies permit. These are not speculative musings—they are necessary inquiries, because compression, secure storage, and real-time analytics are no longer optional. Approaches such as topological embeddings, quantum-resistant data models, and ultra-efficient vector representations are not without precedent. Comparable methods have been deployed in genomics, defense telemetry, and neural time-series forecasting.

Salphaware does not claim all methods are production-ready—but we are production-ready in our discipline to ask the right questions. We follow Rust-based, zero-trust principles not as aspirational optics, but as requirements for infrastructure-grade design. National security-grade architecture demands anticipation of what comes next—not the extension of what is familiar. Our posture is less speculative than grounded in anticipatory realism.

National security is not only about borders. It is about dignity. It is about whether patients and providers can operate with confidence that their system was designed to protect them—not expose them. It is about whether the most private truths in American life—our illnesses, our histories, our families—are governed as sacred trust, not as profit opportunity.

Let us build the architecture worthy of that truth.

E. Federated Public Health Infrastructure: A Coordinated National Defense Layer

In the State of Maryland, a national prototype already exists.

The Maryland Institute for Emergency Medical Services Systems (MIEMSS) is an independent state agency responsible for the full operational command of emergency medical services statewide. Created by statute in 1973, MIEMSS oversees trauma and specialty center designation, statewide EMS licensure and certification, data collection

and analytics, and the command of Maryland's emergency communications and medevac dispatch systems. Through its Emergency Medical Resource Center (EMRC) and SYSCOM center, MIEMSS coordinates real-time patient distribution, hospital diversion, and air-medical transport—ensuring the right patient reaches the right hospital at the right time, with precision that spans roadside response to trauma bay arrival.

In FY2023 alone, EMRC handled over 187,000 clinical calls; SYSCOM managed more than 22,000 telephone consults and 14,000 radio dispatches. MIEMSS also operates the CHATS alert system to coordinate hospital surge and diversion status statewide. This is not conceptual infrastructure—it is real, operational, and replicable.

What makes MIEMSS exemplary is not its size but its structure. Data flows securely. Standards are enforced. Real-time intelligence bridges EMS, hospitals, dispatch, and public health responders. MIEMSS does not replace local systems—it synchronizes them. And in doing so, it offers a living proof of how the federal government might support a broader federated public health architecture without imposing a brittle, one-size-fits-all monolith. MIEMSS demonstrates that coordinated, real-time public health infrastructure is achievable. But it is not yet national *[PA-1]*.

Today, no federal agency—neither the Centers for Disease Control and Prevention (CDC), CMS, nor the Department of Health and Human Services (HHS)—maintains a real-time dashboard of American health system readiness. No agency coordinates interstate hospital diversion, medevac deployment, or cross-jurisdictional public health response. When pandemic waves surge, bioterror threats arise, or mass casualty events overwhelm local systems, the United States relies on static data, patchwork reporting, and improvised coordination. The result is not just inefficiency—it is fragility. And COVID-19 proved, with tragic clarity, that public health in America is structurally uncoordinated.

It is time to make a choice. It is time to choose that America's health is meaningful.

A federated public health infrastructure can unify these fractured capacities into a coordinated national defense layer—preserving state autonomy while empowering federal action. Under this model, the CDC would retain oversight of disease monitoring and containment. CMS would manage resource coordination across Medicare and Medicaid networks. HHS could govern surge mobilization, pharmaceutical caches, and human capital redeployment. The Department of Defense (DoD), the National Guard, and the Department of Veterans Affairs (VA) would become agile partners in deploying equipment, supplies, and clinicians when local resources collapse. FEMA, already tasked with federal disaster response, would gain real-time visibility into health system strain and resource availability—enabling faster, better-coordinated action. The DoD could further leverage this model as a proof-of-concept for coordinated military medical response abroad, while the VA could harness it to maximize veteran care access and outcomes across the full geographic and acuity spectrum, in every community. Public health no longer operates in silos—it becomes a nationally orchestrated system that makes every American a direct benefactor *[TD-2]*.

This model also unlocks extraordinary fiscal and operational efficiency. With centralized purchasing power and shared logistics, the federal government can drive down costs for pharmaceuticals, PPE, diagnostic supplies, and life-sustaining devices—eliminating duplicative procurement cycles across agencies. Just as the Defense Logistics Agency ensures cost control and readiness for the military, a federated health infrastructure can ensure critical supplies reach where they are needed most—swiftly, safely, and at scale.

Yet the value of this architecture extends far beyond disaster scenarios. It enables upstream tracking and downstream action on emerging threats—food and water insecurity, environmental exposure, or supply chain breakdowns that signal systemic health risk. Aligned with geospatial intelligence, the system transforms social determinants of health (SDoH) from academic constructs into real-time, fundable triggers—visible, measurable, and correctable.

It also enables the delivery of highly specialized services—disability-focused care coordination, rare disease surveillance, pediatric specialty routing—ensuring that even the most medically complex individuals receive continuity of care across jurisdiction, payer, and circumstance.

This is far from just an alert system. It is a living coordination framework—capable of powering mobile crisis response, home care services, pharmacy routing, and more. With all ancillary services branching from a shared infrastructure, we gain clarity, cohesion, and control. Every American benefits from a system designed not just to react—but to anticipate, connect, and care.

At the core of this vision lies the National Health Data Repository (NHDR). With NHCI/NHEI-based data architecture and permissioned interfaces for certified public health actors, this system enables cross-agency, cross-state surveillance and action in real time. Local salmonella alerts, waterborne illness spikes, opioid overdoses—lifted from fragmented EHRs to a federated alerting system capable of triggering epidemiologic analysis and just-in-time response *[TD-13]*. Specialty registries—from maternal health to antimicrobial resistance—monitored and resourced dynamically in coordination with live capacity data.

Boots-on-the-ground response becomes intelligent and coordinated. EMS teams in Georgia could receive AI-guided routing from national surge analytics. A hospital in Kansas could be alerted to a respiratory spike modeled by CDC algorithms. A dialysis center in California could auto-request reserve cartridges from federal stockpiles, routed based on real-time risk data. National Guard deployments triggered by fire, flood, or pathogen would no longer be bureaucratic scrambles, but planned escalations—driven by data, not desperation.

This is not just information sharing—it is operational coordination, systematized at scale, where data becomes direction, awareness becomes escalation, and public health moves at the velocity of need.

Salphaware envisions a future in which AI developed at the federal level becomes deployable public health infrastructure—guiding triage, detecting threats early, adapting in real time. Local agencies are no longer expected to invent these tools—they inherit them. National dashboards—scaled from MIEMSS—become standard gear for health departments, EMS systems, and hospital emergency managers across the country.

This design does not centralize every public health action. It establishes shared architecture, atomic-level interoperability, and permissioned roles for action. State and tribal agencies act with autonomy—but not in isolation. Federal actors no longer fly blind. And the American people no longer wonder who is in charge.

This is a national security imperative—and a blueprint for civil resilience.

A robust federated layer could monitor and mobilize beyond traditional healthcare: national pharmaceutical reserves, ventilator inventories, mobile clinics to disaster zones, pre-positioned naloxone or insulin based on forecasted need. It could route behavioral health responders to zones of escalating violence and despair. It could predict maternal mortality by ZIP code or vaccine underutilization by language group.

Public health, finally, would become visible—in real time, on the ground, with national coordination behind it.

No one questions the federal government's role in defending our borders, skies, or seas. It is time we apply that coordination to the defense of our collective health.

Let this be the architecture that makes it possible. Let this be the year the United States becomes capable of seeing its public health threats before they overwhelm us, and responding with a precision and solidarity that only infrastructure-grade design can provide. Let the reference architecture unite certified state, tribal, territorial, and local systems under a common protocol stack. Let public health data transmission be secure, identity-anchored, and real-time—from community dashboards to federal command. Let MIEMSS be the proof of concept. Let the nation become the system—where preparedness is not improvised, but orchestrated; where public health is not peripheral, but foundational; where standards reduce burden instead of creating it, and where every community, from the most rural to the most resourced, becomes part of a unified American shield *[VB-13]*.

F. Federated Portals for Patients, Providers, Government

Portals today are a burden. They fragment care, silo communication, and outsource administrative responsibility to the very people the system is supposed to support. Patients juggle passwords, toggle between disjointed interfaces, and are expected to recall details better stored in the system itself. Providers navigate multiple EHRs, partial records, and disconnected messages—spending time reconciling the irreconcilable instead of delivering care. Public health actors face delays, blind spots, and bureaucratic hurdles that fracture real-time response.

The health world is scrambling to build more and more portals. But based on the merits of the Trilateral Value Proposition, Salphaware rejects that route—one that serves institutions, not patients. Patients do not need more portals; they need fewer. In our vision, they will need only one. A federated portal infrastructure unravels the existing bind.

The National Health Data Repository (NHDR) built on an atomic health data architecture creates not only the backend for a smarter health system—but the frontend for a transformed user experience. Federated portals are the interface.

Under this model, each stakeholder class—patients, providers, public health officials, social services, and government regulators—receives secure, certified portal access to the same longitudinal dataset, governed by role-based permissions and identity-anchored authentication. No more workarounds. No more delayed access. No more siloed logins. The system adapts to its users—not the other way around *[TD-1]*.

For patients, the change is revolutionary. A single login—secured by password, biometric, or facial recognition—would reveal a complete, longitudinal view of their health: every hospitalization, lab, prescription, immunization, scanned document, and payer interaction, regardless of where care was received or which EHR was used *[PC-2]*. Behind this simplicity lies a hardened security framework: encrypted at rest and in transit, identity-verified through NIST 800-63-3 digital standards, and monitored through zero-trust architecture to detect anomalies and prevent misuse. This is continuity made real. Patients would no longer juggle six portals, three passwords, and an Excel sheet of logins just to survive. Instead, they would move through life with a system that remembers, integrates, and supports.

Patients could:

- View all encounters, labs, medications, allergies, and diagnoses in one timeline.
- Share records across systems instantly—with control over who sees what and for how long.
- Schedule care based on earliest availability across all in-network systems within a selected radius and service type.
- View and select medication providers for their prescribed drug by cost.
- Resolve prior authorization in real-time, before care is delayed or denied.
- View billable services before they occur—and know if, how, and where they will be covered.
- Receive telehealth services seamlessly across jurisdictions or vendors.
- Manage chronic conditions with AI-curated nudges, refill alerts, personalized coaching, and community referrals *[PC-1]*.
- Engage with culturally tailored content, SDoH support, and localized services—ride shares, food insecurity, and so much more—flagged automatically through geospatial and clinical data integration *[PC-6]*.

- Receive immediate alerts when results from a community-based lab indicate urgent follow-up—prompting direct outreach from providers or clear instructions to seek care, call 911, or visit an emergency facility.
- Integrate bilaterally with phone apps and wearable devices in real time—where, for example, a smartwatch detects bradycardia, the portal receives the signal, the patient's cardiologist receives a notification, and the patient is alerted with tailored guidance, care escalation options, or monitoring support.

The possibilities are vast—because the architecture is universal. With a federated portal, high-quality virtual mental health care cascades across the nation. A troubled teen logs in from a school counselor's office—or initiates support alone, from a phone—and receives crisis intervention and suicide prevention in real time. Emotional healing for anyone becomes immediate, human, and accessible, powered by qualified professionals and guided by patient need, not system availability.

This same portal becomes the convergence point for provider-prescribed apps—monitoring tools, therapeutic programs, and guided self-help modules—customized to an individual's condition and context *[PC-3]*. Every application is certified through a Trilateral Value Proposition (TVP) lens, rated not by marketing claims but by independently measured outcomes *[PC-5]*. They compete on their clinical impact, ease of provider oversight, and seamless integration with a patient's broader care plan.

Imagine: a diet and activity tracker for obesity; a gamified walking app that simulates Easter egg hunts through a nursing home's garden; a blood sugar and diet app for diabetes; a mood regulation tool for depression—all feeding into a single, unified provider view. A new tab appears on the patient's chart: "Wellness & Participation." It shows gym attendance, community volunteering, and app engagement—not as extraneous data, but as clinical relevance made visible. Social and emotional health are no longer afterthoughts. Care becomes a co-authored story, owned together—where patients and providers navigate health side by side.

Plus, the nightmare of disaggregated scheduling for medical visits and appointments becomes a dream in a federated scheduling architecture *[PC-12]*. Providers and institutions manage their availability *at the provider level* in a federated scheduling tool, and payer- and needs-aware scheduling connects patient demand with the proximal region-level supply of institution-agnostic provider availability coordinated by intelligent algorithms and filters that ensure that the right patient receives exactly the right specialized care at exactly the right time. Providers' license numbers would be the provider-level ID, perfect for scheduling in conjunction with other resources and equipment, such as an MRI machine as a facility-level resource. Optimized booking enhances provider revenue. Single site booking makes the patient's life easier. Schedule confirmations, reschedules, and reminders become automated, familiar, and consolidated. Patients' entire health schedule across specialties is consolidated in a manner that is not only reduced to a single calendar, but also one that easily interfaces with other personal and family calendar apps via a connection that ensures changes are updated in real time.

For providers, the federated portal is not another screen—it is clinical intuition, extended. A portal becomes a diagnostic partner, coordination engine, and patient advocate in this context. Whether used by a primary care physician, triage nurse, pharmacist, or community health worker, the portal offers a structured, complete view of the patient—across locations, disciplines, and time. It surfaces patterns that once lived only in obscurity: deteriorating mood paired with missed follow-ups; spikes in blood sugar following skipped meals; physical isolation reflected in disengagement from app-based walking prompts. It becomes a living dashboard of the patient's narrative between visits—curated by technology, informed by behavior, and actionable in real time.

Providers could:

- Rapidly assess patient history, allergies, medications, and active diagnoses—regardless of prior care location.
- Eliminate the burden of medication reconciliation at admission using pre-verified longitudinal data.
- Refer patients directly to specialists or diagnostics through real-time scheduling with automatic data-sharing.
- Communicate with payers to verify benefits and secure approvals before services are rendered *[PR-4]*.
- Collaborate across disciplines with shared care plans, alerts, and threaded documentation.
- Initiate follow-up for critical test results, missed medications, or escalating risks.
- Leverage embedded AI tools for diagnostic assistance, including image interpretation, sepsis alerts, and genomics-based predictions.
- Fulfill regulatory and quality reporting requirements directly within the care record.

Through this design:

- A pediatrician can portal a parent to coordinate follow-up.
- A public health official can portal a patient after a communicable disease report.
- A social worker can portal a patient to schedule transportation or arrange home food delivery after discharge.
- A pharmacist can portal a patient to address unclaimed prescriptions before harm occurs.
- A community provider can access federal telehealth routing, disability-specific support menus, or interpreter services on demand.

Instead of starting over at every appointment, clinicians log in to continuity. Chasing documentation is replaced by responding to insight. This is where care gaps reveal themselves not through audits, but through ambient data.

For public health and government, federated portals become force multipliers. Local, state, tribal, territorial, and federal agencies gain role-based access to the data they need—without waiting for quarterly extracts or improvising during crisis. The CDC,

CMS, ASPR, FEMA, and U.S. Department of Veterans Affairs could each operate with live visibility into health system readiness, resource allocation, and public health risk—backed by predictive analytics and cross-agency coordination *[TD-17]*.

Agencies could:

- Monitor maternal mortality, overdoses, or antimicrobial resistance in real-time by ZIP code.
- Trigger alerts when asthma spikes correlate with wildfire or air quality events.
- Allocate insulin, naloxone, or ventilators before stockouts occur—based on modeled demand and geography.
- Enable tribal health authorities to co-manage care with the Indian Health Service (IHS) and public health surveillance tools tailored to sovereign needs and cultural protocols.
- Route FEMA support to dialysis centers or nursing homes at risk from water contamination or supply disruption.
- Flag food pantry connections and nutritional support services for flagged patients in rural health centers.
- Integrate veteran care pathways via VA-enabled portals embedded in community care systems.

For public health, the federated portal becomes a precision-grade lens into population well-being. It captures localized surges in respiratory illness before ER volume spikes. It highlights behavioral withdrawal in high-risk youth before it escalates. It maps engagement patterns across age, region, or diagnosis—transforming portals from passive interfaces into national biosurveillance tools. When anonymized and aggregated, these data streams offer real-time insights into population trends, social determinants, care access, and intervention success. Public dashboards can reflect the health of a county, the resilience of a school district, or the equity of outcomes across lines that once remained invisible. It is where national health policy gains traction not through mandates, but through embedded measurement—automated, auditable, and impossible to ignore.

This is where access is no longer fragmented, coordination is no longer improvised, and every American engages with a single portal that protects, clarifies, and delivers.

Let this be the front door to the health system the nation deserves. Where infrastructure meets experience. Where citizens no longer regularly reconstruct their lives in intake forms. Where rural and tribal clinics operate with the same precision as academic health centers. Where insight becomes action—without delay.

And from this portal, the nation itself can unite—not just in care, but in shared experience. Imagine coast-to-coast 5Ks promoting cardiovascular health. USDA-aligned gardening challenges staged by harvest season and climate zone to support food pantries. Virtual senior meetups by hobby and language. Naturopathic wellness programs where diet, herbs, and culture coalesce into personalized care.

This is not just health intervention. It is infrastructure of the people, by the people, and for the people—reimagined as social technology, with guardrails for purposeful engagement and public good.

A single federated portal is the broker and the key. In an irony worthy of celebration, it is technology that returns U.S. medicine and its citizenry to its humanity. These are civic bridges—moments of connection in a time when division is the easier path. Through health, America unites. Through health, America becomes whole again.

G. Unleashing the Health Technology Market Economy

The transformation proposed in this architecture is not merely regulatory or technical—it is catalytic. It will unleash the largest coordinated health system modernization in modern U.S. history. With it comes the most consequential economic opportunity in the saga of American healthcare.

By standardizing identity through NHCI/NHEI, unifying data through atomic architecture, and establishing the National Health Data Repository (NHDR) as the singular source of truth, this proposal breaks open an ossified system and sets loose a cascade of innovation, competition, and entrepreneurship across every layer of the health economy.

A Richter-Scale Economic Awakening

The moment health data becomes shared, standardized, and secured, the U.S. exits its closed-system adolescence and enters the post-EHR, cloud-native, post-quantum era of health innovation. Interoperability is no longer trapped behind EHR vendor walls. Data becomes fluid—and with it, the marketplace explodes.

This is not sectoral reform. It is a multi-industry reanimation:

- Technology vendors—from wearable device makers to advanced diagnostics platforms—gain real-time access to standardized, permissioned datasets, opening the door to a universe of consumer-grade tools with clinical-grade insight.
- Cloud and cybersecurity companies compete to protect post-quantum infrastructure, advancing zero-trust enforcement and intelligence-grade verification.
- Interface designers disrupt the market for provider and patient engagement, replacing dated, fragmented portals with elegant, AI-augmented interfaces custom-tailored to role and task.
- Public health infrastructure firms are commissioned to design and construct MIEMSS-like regional coordination hubs—brick-and-mortar centers of surveillance, triage, and multi-agency command.
- Consulting and IT transformation firms surge to support regionalization, certification, and workforce modernization.

- Quantum computing companies gain direct application in federated learning, quantum key exchange, and secure analytics—with a funding and policy environment that invites commercialization.
- Biomedical logistics and bulk procurement firms emerge to manage shared national inventories, coordinate public-private acquisition contracts, and optimize pharmaceutical distribution with AI-driven precision.

New Jobs, New Firms, New Industries

Entire job categories will emerge—roles never before seen in American healthcare, yet soon indispensable to its daily operations. These will not be fringe specialties, but foundational professions within a modernized system—designed to manage what legacy infrastructure could not: real-time data, distributed intelligence, and cross-agency orchestration. From quantum-secure system architects to AI-augmented triage technicians, this is not a speculative vision. It is the next normal:

- Clinical AI designers
- Public health informaticians
- Federated system auditors
- Biometric portal engineers
- Telehealth coordination brokers
- Real-time appeals agents
- AI safety and bias officers
- Atomic data compliance analysts
- Federated procurement strategists
- Biomedical supply-chain modelers
- Genomic risk interpreters
- AI-augmented triage technicians
- Secure telehealth logistics engineers
- Health equity surveillance analysts
- Virtual care integration liaisons
- Quantum health system architects
- Resilience funding analysts

Every layer of the proposed architecture—policy, implementation, interface—generates its own economic substructure.

A New Economic Framework for Old Problems

Where the legacy system imposed artificial scarcity and technical roadblocks, this architecture eliminates constraints and creates new pathways for fiscal responsibility, efficiency, and service optimization.

Hospitals gain:

- Breach risk offload: Long-term data retention liability is shifted to the NHDR.

- Cost reduction: Removes redundant storage, manual reconciliation, and compliance overhead.
- Revenue optimization: Automates audit trails and appeals generation using defensible, timestamped data provenance.
- Labor reallocation: Staff previously dedicated to abstraction and compliance can be redeployed to frontline operations and innovation.

Patients gain:

- Unified control: NHCI gives every person a single point of command over their entire health history and its permitted uses—unlocking participation in data-sharing economies, enabling personalized services, and empowering individuals to leverage their data as an asset in value-based care, clinical research, or wellness markets. As individuals gain control, they become active participants—not just subjects—in a health data economy that rewards transparency, incentivizes participation, and invites consumer-choice ecosystems to flourish around their needs.
- Service optionality: With EHR data oligopoly broken, patients can choose among competing apps, virtual clinics, diagnostics tools, and AI-based care managers.
- Price and access visibility: Patients know where care is affordable, covered, and high-quality—before it is delivered *[TD-19]*.

Vendors gain:

- Open ecosystem access: No longer dependent on bespoke EHR contracts, developers build for the NHDR, instantly unlocking nationwide potential *[PR-12, TD-1]*.
- Accelerated training: Secure, permissioned longitudinal datasets enable faster, safer AI model development.
- Lower market entry barriers: A single certified connection to the data layer replaces years of EHR integration work.

Payers gain:

- Fraud prevention: Cryptographic timestamping and atomic-level traceability eliminate duplicate claims and hidden manipulation.
- Real-time prior auth: Requests and decisions are brokered across payers and providers before care is delivered—not weeks after.
- Market competition: Cost transparency forces competition on value and experience, not opacity.

Supply chains gain:

- Bulk purchasing leverage: Centralized demand visibility drives down drug, equipment, and PPE costs via consolidated procurement.

- Forecasting and logistics: Portal-based ordering predicts demand and flags supply risks in advance of stockouts.
- Geo-optimized distribution: Orders are placed locally but guided by national intelligence—no more surplus in one state while another suffers.

The Downstream Economy

These reforms do not simply affect the existing value chain. They spawn new ones:

- A telehealth platform integrates AI triage and genomic prediction, routing patients in real time based on risk, availability, and benefit design.
- A rural health center bulk-orders diagnostic kits via portal-based procurement, triggered by risk flags in the NHDR and shipped via the lowest-cost national partner.
- A tribal health authority in Arizona builds a culturally tailored portal module that supports native languages and sovereign health protocols. It integrates with IHS systems, federal surge indicators, and then licenses the model to other tribal systems—turning policy architecture into local economic opportunity.
- A start-up incubator in Cleveland builds three dozen NHDR-native tools—supporting maternal health, stroke recovery, behavioral health, medication adherence—each ready for nationwide adoption.

And in Crisis, This Economy Accelerates

When a new pandemic wave, cyberattack, or bioterrorism incident arises, this infrastructure does not stall. It accelerates capital. It routes aid. It connects coordination with funding. Infrastructure investments made in peacetime become triggers for immediate reimbursement, deployment, and mobilization. Contracts no longer take months—they are activated in hours. Crisis readiness becomes a profitable vertical. Supply stockpiles and surge services become part of the recurring national budget, fueling growth not just in recovery, but in resilience.

Let the Market Build What the Architecture Makes Possible

This is not about controlling innovation. It is about unleashing it. The government builds the scaffolding. The private sector populates it with value.

Let this be the digital health equivalent of the interstate highway system—not paved in asphalt, but in code, trust, and opportunity. A national backbone of information where healthcare is not just delivered, but optimized, commoditized, and exchanged—with the velocity of trade and the precision of care.

Let this be the boldest reallocation of federal healthcare investment since Medicare—not as a safety net, but as an engine. One that turns stagnant infrastructure into economic velocity, market innovation, and national resilience.

Let this be the moment the United States of America turns health data from a liability into a launchpad—where patients no longer chase care, providers no longer guess in the dark, and entrepreneurs no longer build in walled gardens.

Let it not be delayed by incumbents.
Let it not be diluted by compromise.
Let it begin—because the future of American health, security, and prosperity is not a cost center. It is the next immense growth center**.** And it is ours to build.

IV. Empowering the Existing Ecosystem: QIN-QIOs and QHINs as Catalysts

This section addresses the following CMS RFI questions:
PC-10, PC-11, PC-14, PR-7, PR-11, TD-1, TD-13, VB-8, VB-11

A. Positioning QIN-QIOs as Stewards of National Health System Transformation

QIN-QIOs have long served as trusted extensions of CMS, advancing clinical quality and reducing disparities across the U.S. health system. As defined under 42 CFR § 475.101, their core responsibilities—technical assistance, clinical data enablement, and regional quality improvement—make them uniquely qualified to anchor major aspects of the transformation envisioned in this response. Their mandates under the QIN-QIO program, as scoped by the 13th SOW Task Order Request for Proposal (TORP), provide foundational capacity and geographic reach that can be leveraged to expand their system impact.

In the re-architected ecosystem described in this response, the role of QIN-QIOs becomes not smaller, but profoundly larger. As the healthcare system transitions into a modern, intelligent, interoperable data fabric, QIN-QIOs are natural levers to steward that transition while preserving system stability. Their function expands from deployment partners to systemic coordinators, policy accelerators, implementation auditors, regional educators, and guardians of continuity.

New Frontiers of QIN-QIO Functionality in the National Data Infrastructure:

- Digital Change Management: Help providers navigate the transition from legacy mechanisms to intelligent platforms—including coordination of trainings, user buy-in, and staged implementation support.
- Quality Oversight and Technical Auditing: Act as regional validators for certified data pipelines, ensuring fidelity of compliance architectures and safeguarding patient protections.
- Support for Academic Linkages: Broker partnerships with academic institutions to align curricula, training programs, and research agendas with the forward trajectory of national digital health infrastructure.
- Disaster Preparedness and Data Resilience: As continuity-of-care becomes a national security issue, QIN-QIOs can design and deploy response protocols, simulation exercises, and backup systems in alignment with CMS and federal preparedness objectives.

- Extension of Care Across Continuums: QIN-QIOs will continue serving nursing homes, dialysis centers, and rural providers—using modular technology add-ons that help these facilities engage with the national platform without requiring system overhauls.
- Accountability Layer for Commercial Actors: Ensure EHR vendors, health IT platforms, and regional aggregators conform to CMS' equity and transparency principles—serving as truth-tellers and enforcers of compliance and access.

Importantly, QIN-QIOs can perform these roles while preserving—and even enhancing—their current portfolio of services. The complexity and breadth of their mission will necessarily expand to meet the operational complexity of a reengineered national health infrastructure. The scale and pace of implementation envisioned in this RFI will generate uncertainty, disruption, and high stakes for local providers. QIN-QIOs are CMS' first line of support in easing that transition *[PR-7]*.

While Salphaware builds the regulatory-grade infrastructure and automated tooling that powers this new landscape, explored further in Section VI, QIN-QIOs ensure it functions with equity, trust, and fidelity across all geographies and care settings.

To preserve innovation velocity and product integrity, we recommend CMS formally encourage partnerships between QIN-QIOs and agile private-sector actors like Salphaware. This model:

- Protects IP and time-to-market by avoiding government-driven reinvention.
- Multiplies CMS reach by giving QIN-QIOs field-tested tools.
- Enables QIN-QIOs to deliver value at scale without duplicating technical development or federal R&D effort.

CMS may also consider funding QIN-QIO participation in early-phase pilots via cooperative agreements or regional implementation grants—ensuring the transition to modern infrastructure is supported by trusted local partners *[TD-1]*.

Finally, we recommend CMS consider whether QIN-QIOs, with their national footprint and mission-aligned mandate, could serve as regional oversight partners during the rollout of universal identity systems (NHCI, NHEI), including consumer-facing credential adoption, compliance auditing of FHIR data flows, and public engagement on consent and transparency *[PC-14, TD-13]*.

In the future ecosystem, the role of QIN-QIOs is not just preserved—it is amplified. They will continue enabling clinicians. But they will also train tomorrow's workforce, shepherd multi-system transformation, and serve as the connective tissue between innovation, equity, and operational excellence. They are CMS's enduring stewards of quality, equity, and continuity—now rearmed for a digitized, data-first American health system *[VB-11]*.

B. Redefining QHINs: Value Through Intelligence, Not Just Interchange

The Trusted Exchange Framework and Common Agreement (TEFCA) is a monumental step toward national data liquidity *[PC-10]*—but interoperability alone does not guarantee improvement. Data must be more than exchanged; it must be transformed into something actionable, trustworthy, and meaningful. QHINs are central to that vision, but their long-term value will depend on evolution: from passive routers of information to active enablers of clinical intelligence and regulatory precision *[PC-11]*.

As CMS transitions toward a centralized data infrastructure—such as the National Health Data Repository (NHDR), the National Health Citizen ID (NHCI), and the National Health Entity ID (NHEI)—QHINs will no longer be judged by their ability to simply move data. They will be evaluated based on their ability to ensure that the data moved is clean, structured, certifiable, and usable across complex stakeholder demands. Salphaware's compliance and abstraction infrastructure supports that evolution, offering intelligence layers that are complementary to QHIN routing, not competitive with it.

We propose a model where QHINs deliver value not through ownership of data, but through what they enable with it. This model respects TEFCA's principles while accelerating its practical impact across clinical, regulatory, and public health domains. Specifically, QHINs can become force multipliers for:

- Regulatory signal generation: ensuring TEFCA-based exchanges produce structured datasets that satisfy measure abstraction, appeals documentation, and audit trails;
- Data quality assurance: embedding hygiene protocols and compliance metadata within the flow itself;
- Infrastructure trust: serving as regional validators for national frameworks like the NHDR, NHCI, and NHEI *[PR-11]*.

Salphaware is currently focused on hospital-grade compliance and reporting services, while actively aligning its infrastructure with QHIN-compatible standards. Our goal is to serve as a value-tier contributor within the national exchange framework—whether through future QHIN designation, strategic partnerships, or federal interoperability expansion.

Salphaware and QHINs are not duplicative in scope—but operate adjacently, as complementary actors converging on a shared goal. Together, we envision a future where:

- QHINs facilitate secure, trusted routing;
- Salphaware automates regulatory-grade interpretation and transformation;
- Hospitals, payers, and CMS benefit from intelligent, standards-based interoperability—without human rework or duplicate effort.

To help realize that future, we encourage CMS to:

- Formalize a designation pathway for purpose-driven QHIN partners—those focused on delivering measurable downstream value from data, not just data movement;
- Incentivize QHIN + private-sector partnerships with actors like Salphaware that embed modular logic within TEFCA-aligned networks;
- Avoid vendor lock-in by prioritizing modularity, local adaptability, and ecosystem extensibility;
- Promote a QHIN evolution model rooted in the Trilateral Value Proposition, where QHIN activities demonstrably benefit patients (e.g., continuity of care), providers (e.g., reduced rework), and government (e.g., compliance-ready datasets) *[VB-8]*.

These partnerships should prioritize interoperability, modularity, and local adaptability—ensuring that transformation logic from actors like Salphaware can be embedded flexibly across different QHINs or state-led networks without limiting stakeholder choice.

Salphaware believes this model—QHINs as trust-bearing routers and regulatory-intelligence enablers—can be piloted and matured within the next three years. As CMS begins to scale the NHDR vision and related national frameworks, QHINs that adopt this dual-function model will be well-positioned to support continuity, compliance, and innovation simultaneously.

In that ecosystem, QHINs remain critical—but only if they evolve beyond passive transport. They must embrace their role as connective, intelligent, compliance-conscious infrastructure—not by becoming everything, but by partnering with those who build what they were never designed to create.

We are not building a parallel system—we are building the logic layer that activates national infrastructure, transforming exchange into interpretation and movement into meaning.

V. Clinical, Technical, and Policy Integration to Prevent Avoidable Harm

This section addresses the following CMS RFI questions:
PR-2, PA-1, VB-13

SEP-1 is not merely a measure—it is a regulatory skeleton around which patient safety, equity, and accountability coalesce. Its automation is not a burden—it is a release. It is a platform for national learning, equity tracking, and litigation defense, if built right.

A. Unified Sepsis Standard: Reducing Mortality, Denials, and Data-Driven Fraud

The United States cannot afford to operate a healthcare system where patients are measured by one standard, paid for under another, and denied care by a third. Yet that is precisely what has emerged in sepsis care: CMS rightfully uses SEP-1 to measure patient safety performance, with underlying clinical logic historically aligned to Sepsis-2. Hospitals apply this logic in both care delivery and ICD-10-based coding for reimbursement. But commercial payers—including Medicare Advantage plans—exploit the more restrictive Sepsis-3 framework to deny or downgrade claims, asserting that patients never met criteria despite receiving timely, life-saving care.

This is not an administrative inconvenience—it is a lethal misalignment of definitions, executed through technology and payment logic that serve payers' financial interests while exposing hospitals to unrelenting clinical and financial threat.

Sepsis-3 Is Logically Broken and Dangerous by Design

Sepsis-3's definitional flaws are both logically and scientifically incoherent:

- It embeds intervention-dependent criteria—such as the administration of vasopressors—into the diagnosis itself. In practice, this means the sepsis diagnosis is only "confirmed" after the patient has deteriorated enough to require aggressive treatment. This cart-before-the-horse diagnostic model is circular logic by definition—a logical fallacy—which disables early recognition and requires failure to validate itself.
- Sequential Organ Failure Assessment (SOFA) is a well-studied measure of mortality. The authors of Sepsis-3 crudely grafted SOFA as a measure for sepsis, and there is no shortage of clinical research opposed to this approach. This commits a transitive property fallacy by positioning SOFA scores as both a diagnostic marker and an outcome proxy. If SOFA = sepsis, and SOFA = mortality, then by implication, sepsis = death—a false equivalency that distorts both epidemiology and resource allocation. It embeds the fatal outcome into the

definition, rather than enabling clinicians to prevent it, and conceptually represents a dangerous proximity to a confounding variable in this context.

Even the authors of Sepsis-3 warned in the research itself, *"Neither SOFA nor qSOFA are intended to be a stand-alone definition of sepsis."* Yet commercial payers—including many Medicare Advantage plans—routinely use Sepsis-3 to justify DRG downgrades, audit takebacks, and outright denials. HANYS called this out in 2019. AHIMA warned of coding chaos in 2020. OIG launched audits in 2022. Each saw the writing on the wall—clinical harm masquerading as payment policy—but to date, no U.S. regulating body seems to have the upper arm strength to throw the red flag, even when it threatens human life and defrauds hospitals.

Delays in recognition are not benign—sepsis mortality increases by approximately 8% for every hour treatment is delayed. A framework that obscures or delays diagnosis is not just clinically flawed; it is fatal by design. The result is a national bait-and-switch, where hospitals are graded by a meaningful measure and reimbursed under a false one.

The Salphaware Response: Truth Through Technology

What we are witnessing across the nation is the misuse of health IT—not as a harmonizer of clinical truth, but as an instrument of definitional disparity. Technology is being used to automate denial logic, rather than to align care with evidence, payment with regulation, and measurement with reality. This is the moral hazard of unregulated interoperability—where, in its current form, definitional manipulation enables a systematic transfer of funds from providers to payers *[PR-2]*, bordering on institutionalized fraud, and subsidized in part by the very government charged with preventing it. It must be addressed by aligning technology with a single, certified, clinically valid definition of sepsis.

To reinforce this effort, Salphaware's *Sepsis Studio* proof-of-concept captures every sepsis case across all major clinical and operational definitions in use today—including SEP-1, Sepsis 2, and Sepsis-3—allowing for transparent comparison, appeals support, and definitional research into whether Sepsis-3 has cost patient lives or caused harm in exchange for payer financial advantage. This work will be measurable, data-driven, and built to separate clinical truth from administrative distortion. It is how technology can demystify and de-weaponize criteria ambiguity—and in doing so, protect both patient outcomes and policy integrity.

The Precedent Exists—And I Helped Build It

When this definitional disparity first emerged in New York, I personally initiated the advocacy to the Healthcare Association of New York State (HANYS) that led to the joint Article 44 MCO Guidance issued by the New York State Department of Health (NYSDOH) and Department of Financial Services (NYSDFS). That guidance—the first of its kind—affirmed Sepsis-2, which is aligned to SEP-1, as the state-standard framework for managed care organizations (MCOs). This guidance was supported by

pre-existing clinical-definitions-turned-state-law (10 NYCRR § 405.4(8)(a)). It restored clinical consistency, protected hospitals, and slowed the proliferation of wrongful reimbursement denials.

CMS should now reify this precedent as federal policy—not just in concept, but in function—by requiring a singular, enforceable sepsis standard across all certified APIs, CDS platforms, abstraction tools, and payer adjudication systems. That is the only way to guarantee that what CMS measures is what hospitals deliver—and what payers must honor.

A One Truth Standard for Sepsis

This is what the trilateral value proposition demands:

- Patients should not face denied coverage or downstream cost risk due to arbitrary, retroactive thresholds that fail to reflect clinical judgment. This reflects widespread provider concern raised in peer-reviewed critiques of Sepsis-3's use in reimbursement logic (e.g., Kleinpell et al., *CCM*, 2016).
- Providers should be rewarded—not penalized—for early intervention that prevents deterioration. Reimbursement models must align with timely, patient-centered action.
- Government should stop footing the bill for definitional shell games played by commercial and Medicare Advantage payers. These misalignments allow payers to deny claims that meet federal quality standards—leaving CMS to underwrite the consequences of delayed care, preventable ICU escalation, and eroded provider trust.

We urge CMS to act with clarity and moral force. Mandate a One Truth Standard for sepsis—technologically enforced, scientifically valid, and impervious to financial manipulation. Doing so will save lives, restore fiscal integrity, and unify the most critical quality measure in American healthcare under one clinically valid standard.

Enabling Automation and Compliance with National Infrastructure

To accelerate national uptake of sepsis performance automation—particularly among resource-constrained hospitals *[VB-13]*—CMS should consider awarding provisional full credit toward SEP-1-related domains of the Hospital Value-Based Purchasing (VBP) Program for hospitals that implement and actively use certified SEP-1 automated abstraction and decision support tools.

This provisional scoring period (e.g., 2–3 years) would function as a time-limited performance credit mechanism akin to safe harbor provisions used in other CMS programs. It would reward early adoption of tools that ensure higher-fidelity data capture, align directly with CMS' burden reduction goals, and promote continuous quality improvement. More importantly, automation ensures that performance measurement is

not disproportionately influenced by facility size, staffing capacity, or health IT sophistication—advancing equity in both care and compliance.

Certification could follow a light-touch attestation or ONC-affiliated mechanism to ensure interoperability, auditability, and transparency. CMS has previously deployed similar approaches under Meaningful Use, MIPS Improvement Activities, and EHR-based eCQM transitions. Applied here, such a model would spur voluntary alignment before a full eCQM transition is technically feasible across all providers.

We urge CMS to:

- Mandate *automated* SEP-1 abstraction for all Inpatient Prospective Payment System (IPPS) hospitals;
- Enable automation to meet compliance standards, with future certification pathways supported by CMS.

These actions would be squarely aligned with CMS' ongoing digital quality measurement strategy and broader modernization goals articulated through initiatives like *Patients Over Paperwork* and the *National Quality Strategy*. They will prevent avoidable death, increase cross-hospital consistency, and ensure that no clinician, patient, or hospital is punished for doing the right thing too early.

B. National Pediatric SEP-1 Standard: Technology and Policy to Save Children's Lives

Each year, sepsis claims more pediatric lives in the United States than all childhood cancers combined. Yet, while national attention and funding have rightly supported aggressive standards for pediatric oncology, there remains no federal sepsis performance measure designed specifically for children.

This omission is not simply an oversight. It is a systemic inequity that costs lives.

Hospitals are required to abstract adult SEP-1 cases, but no parallel obligation exists for pediatric patients, despite clinical consensus that early recognition and treatment dramatically improve outcomes across all ages. The absence of a national pediatric sepsis standard means thousands of hospitals operate without a consistent framework for detecting, measuring, and improving pediatric sepsis care *[PA-1]*. The result is a patchwork of variable practice and missed opportunities for life-saving intervention. These omissions contribute to preventable ICU admissions, prolonged hospitalizations, and costly post-acute care—all of which impose financial burdens on CMS and reinforce disparities in pediatric outcomes across rural and under-resourced facilities.

This gap is no longer technically necessary—or ethically defensible.

New York State has already demonstrated proof of concept. In response to the death of Rory Staunton, the NYSDOH implemented the nation's first pediatric sepsis protocol

requirement, backed by operational definitions and reporting infrastructure. The Rory Staunton Foundation continues to advocate for national adoption of that model, supported by both literature and outcomes research. The precedent exists. The infrastructure exists. What remains is national action.

Creating a pediatric SEP-1 standard aligns directly with CMS' Health Equity Strategy and Digital Quality Measurement goals—advancing data modernization while ensuring that children are no longer invisible in national quality measurement frameworks. CMS is encouraged to now do what the science, the ethics, and the evidence require:

- Establish a federal pediatric SEP-1 standard grounded in New York's model;
- Codify it within the Hospital IQR Program to ensure equity in sepsis surveillance and care;
- Mandate automated abstraction of pediatric sepsis cases, ensuring that compliance is possible even for resource-constrained institutions;
- Encourage the development and certification of burden-negative digital tools that translate this measure into practice without adding documentation overhead *[PR-2]*.

By doing so, CMS will not only close a fatal gap in pediatric quality measurement—it will help protect the lives of children nationwide through technology-enabled accountability**.**

This is not a commercial recommendation. It is a moral one.

Equal protection under federal law cannot exclude the youngest and most vulnerable. Pediatric patients deserve the same rigor, visibility, and safety scaffolding already afforded to adults. A pediatric SEP-1 standard would be a powerful equalizer—driven by data, backed by precedent, and made viable through automation. We are overdue to equalize pediatric patient safety nationwide—especially because agents like Salphaware are poised to heed this call in a burden-negative fashion for providers, in alignment with CMS' burden reduction goals, ensuring that equity is not delayed for reasons of cost or complexity *[VB-13]*.

VI. Salphaware's Forward Path in the National Health Ecosystem

This section addresses the following CMS RFI questions:
PR-1, PR-2, PR-8, VB-13, VB-14

Salphaware is a health technology company born from within the trenches of U.S. hospitals and EMS, built to transform compliance and quality improvement from burdensome processes into intelligent, human-supporting systems *[PR-8]*. Our work is anchored in the Trilateral Value Proposition—a standard we originated to ensure every innovation serves patients, providers, and public programs alike *[VB-13]*.

We connect code to regulatory minutiae, clinical nuance, operational realities, and fiscal microcosms—with stakeholder-level understanding. For us, AI is not a novelty—it is right-sized into mainstream.

Soon, health quality and health technology will become productively commingled terms, as the latter increasingly, poignantly, actualizes the former. We are building systems that automate abstraction, documentation, appeals, and compliance *[PR-2]*—functioning silently in the background to allow boots-on-the-ground humans to be redirected to where actual care—not documentation, clicks, or rework—is needed most *[PR-1]*.

You asked for boldness. We heard you. And we delivered. We do not know yet if Salphaware will become the certifier, the builder, the licensor, the QHIN, or the guide. But we know we are holding a vision. Our legacy is already in motion. And we welcome the opportunity to help shape yours *[VB-14]*.

VII. Strategic Recommendations Summary

These recommendations do not merely respond to CMS and ASTP/ONC's Request for Information—they recast the assignment. They define what is possible, what is operationally ready, and what is structurally overdue.

They reflect a unified infrastructure: interoperable by design, intelligent by architecture, and enforceable by policy. Rooted in implementation, align with the CMS Innovation Center's core priorities: equity, accountable partnerships, and data-driven modernization.

Each recommendation maps precisely to stakeholder categories and RFI question codes, as detailed below and substantiated throughout this submission. What follows is not aspirational—it will transform how federal healthcare is delivered, measured, and secured.

CMS Strategic Recommendations Overview

CMS Recommendation	Section	RFI Questions
Adopt the Trilateral Value Proposition (TVP) as a CMS-wide contracting and innovation standard.	II.A	PR-2, PR-8, VB-7
De-risk hospital innovation participation through grants, VBP safe harbors, and CEHRT flexibility.	II.B	TD-1
Prioritize Veteran-Owned Small Businesses (VOSBs) in innovation contracting.	II.C	—
Establish a National Health Identity Framework (NHCI/NHEI) to anchor longitudinal data access and interoperability.	III.A	PC-14, PR-9, TD-3, VB-13
Adopt and certify AHDA as the national data architecture standard to enable extensibility, automation, equity tracking, and AI-ready interoperability.	III.B	PR-7, TD-3, TD-7, TD-13, VB-5, VB-13
Establish the National Health Data Repository (NHDR) as the nation's secure, standards-based data backbone.	III.C	PR-1, PR-7, PR-12, TD-13, TD-17, VB-12, VB-14
Advance beyond FHIR: pilot scalable, secure alternatives and elevate post-quantum programming, protocols, and zero-trust models.	III.D	TD-9, TD-10, TD-15

Federate public health infrastructure using NHDR/NHCI/NHEI to enable MIEMSS-type surveillance and response.	III.E	PA-1, TD-2, TD-13
Establish a single federated portal architecture with secure, identity-verified access for patients, providers, and public health.	III.F	TD-1, PC-2, PC-14, PR-10
Open NHDR to certified third-party tools to break vendor lock-in and accelerate modernization.	III.G	TD-1, PR-12
Expand QIN-QIO roles to lead change management, credential rollout, auditing, and private-sector partnerships.	IV.A	PR-7, PR-11, PC-14, TD-13, VB-11
Redefine QHIN value as structured data delivery, modularity, and downstream impact aligned with TVP.	IV.B	PC-10, PR-11, VB-8
Mandate a "One Truth Standard" for sepsis aligned to SEP-1 across APIs, CDS, abstraction, and payer logic.	V.A	PR-2, PR-7, VB-13
Establish certified automation pathways for SEP-1 abstraction and VBP credit for early adopters.	V.A	PR-2, VB-13
Establish a national pediatric SEP-1 standard and mandate automated abstraction.	V.B	PA-1, PR-2, VB-13
Accelerate real-world deployment of AI-powered abstraction, appeals, and compliance infrastructure through partnerships and certification.	VI	PR-1, PR-8, VB-14

This Is Not a Wishlist. It Is a Deployment Plan.

Salphaware does not merely support the CMS vision—it mirrors it, accelerates it, and enforces it through executable systems design. These recommendations do not offer incremental tweaks. Each component—identity, data architecture, portals, AI, public health infrastructure—has been mapped to implementation pathways that are real, measurable, and compliant.

This is a blueprint for equity—not just in principle, but in enforcement.

It is a modernization framework—not just of what systems do, but how they coordinate, how they protect, and how they deliver value across patient, provider, and government lines.

Salphaware is ready to pilot its piece of this vision now. Let this RFI be the pivot. Let this be the moment CMS shifts from fragmented compliance to intelligent infrastructure.

VIII. Epilogue: One Worth Building

Yes, Salphaware is a for-profit company by design. But our profit model is tethered to a deeper mission: deliver measurable benefit to patients, operational relief to providers, and actionable intelligence for government. We believe this alignment of incentives is not only ethical—it is the course correction of a lifetime in a system long broken.

As a rural volunteer first responder, I first performed CPR at sixteen. Then came the Army, where I trained as a combat medic and learned the weight of duty above all else. Later, as a civilian paramedic, I stood in blood-soaked boots, haunted by memories of avoidable death I'd rather forget. In government, I watched unfunded mandates trickle down as pain multipliers for the very providers they were supposed to help. Then came the big one: as a hospital sepsis coordinator, I sat in peer review committees, performed audits, and wrote appeal letters for care rightfully given and wrongfully denied. And the young mom who died from septic shock because the signs were missed—she is part of this. How I wished there had been a way to know and nudge the provider.

Now, as a principal scientist in AI and machine learning, I see the same pattern repeated: data systems tragically disconnected from the human lives they are meant to serve. And as the founder of a mission-driven health technology company, I clearly see the toxic past we as a nation keep returning to—legacy actors, legacy thinking, and legacy solutions.

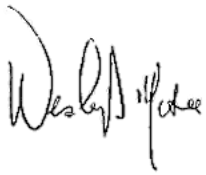
I am also a patient. I had open-heart surgery at thirty-three. I have an ADA-recognized condition and a platoon of specialists—rheumatology, neurology, cardiology, and more. I retell the same story to each new provider. I manage more portals and passwords than any person should ever need. And I keep a ledger of reimbursement receipts just to prove care that already has a digital footprint. I deserve better. My family deserves better. So do millions of others. And so do you.

This vision wasn't born in a conference room. It was sutured together from ambulance bays and audit files, in M.A.S.H. unit training and MRI tables, in denial letters and waiting room clipboards, and in the exhausted eyes of colleagues trying to do the right thing inside a system that makes it hard to try at all. This isn't a pitch. It's not an ask. It's a reckoning.

In a town like D.C., where everyone wants to be a somebody, I am perfectly comfortable knocking on your door as a nobody. This isn't about recognition. And it certainly isn't an appeal for the next contract—we have neither the wealth nor the influence for that expectation. This is simply about putting substance in your hands that is more than me, and more than you. This manifesto isn't mine. It's America's.

And now you just have to be strong enough to make it real.

Respectfully,



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